Towards Localized Surrogate models in Quantum Machine Learning

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Introduction

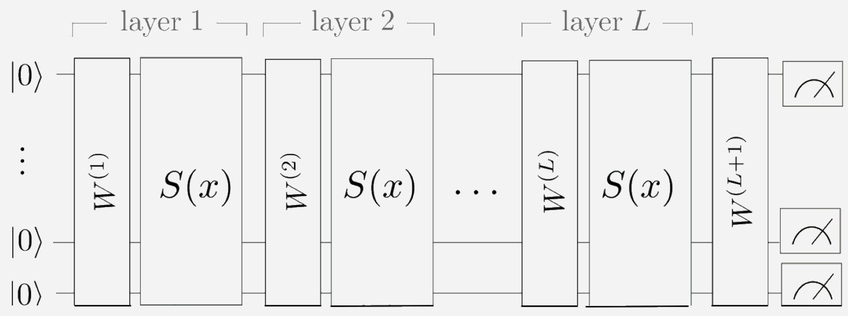
Quantum learning models as Truncated Fourier series:

A univariate quantum model f(x) defined as the expectation value of some observable with respect to a state prepared through variational quantum circuit with parameters θ.

$$f_\theta(x) = \langle 0|U^\dagger(x,\theta) M U(x,\theta)|0\rangle$$

As it consists of L layers each with a data encoding block and a trainable block it is refered to as re-uploading type models. These can be written as a partial Fourier series (Schuld et al (2021)):

$$f(x) = \sum_{n\in\omega} c_n e^{inx}$$



Global Surrogates

Global surrogates for quantum learning models:

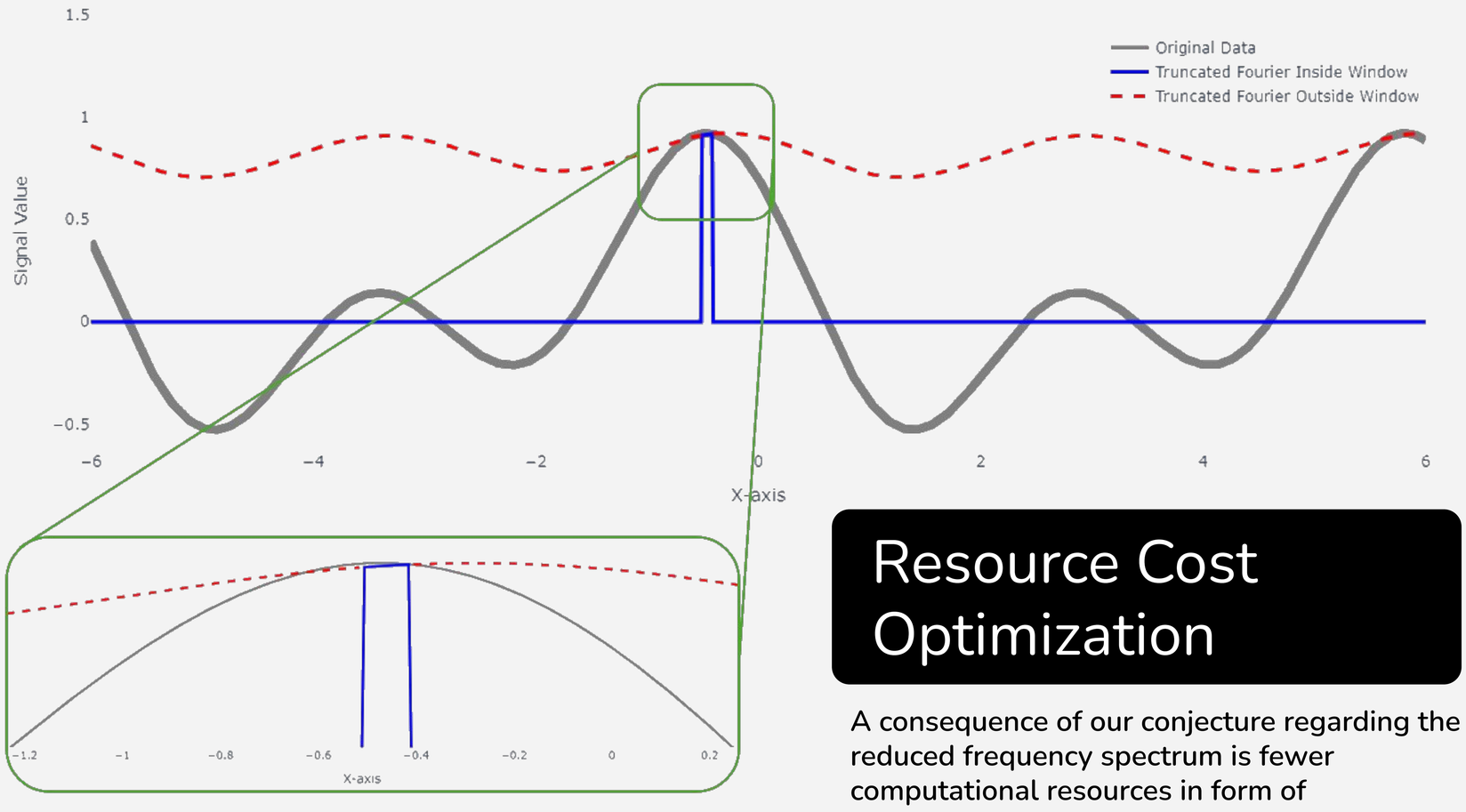
DEF: A hypothesis class of quantum learning models F has classical surrogates if there exists a mapping $S : F \to G$ such that for each $f \in F$, the corresponding classical model, $g_f \in G$ satisfies that the maximal deviation, $|f(x) - g_f(x)|$ is bounded with high probability

$$P\left[\sup_{x\in\mathcal{X}} \|f(x) - g_f(x)\| \leq \epsilon\right] \geq 1 - \delta$$

Fourier based surrogates:
Truncated fourier representation of re-uploading type quantum learning models admit construction of classical Fourier based surrogates for that particular category of learning models. (Schreiber et al (2022))
The number of layers in the re-uploading type model determines the required frequency spectrum for the fourier surrogates.

Windowing

1. In digital signal processing (DSP), windowing is a technique used to isolate a finite segment of a signal by applying a window function, which smoothly tapers the signal to zero at the edges of the segment.
2. Window functions are mathematical functions, such as the Tukey, Hamming, Hanning, or Blackman windows, designed to minimize spectral leakage when performing operations like Fourier transforms on finite signal segments.



Local Surrogates

Proposal:

Local surrogates for quantum learning models

DEF: A hypothesis class of quantum learning models F has classical surrogates if there exists a mapping $S : F \to G$ such that for each $f \in F$, the corresponding classical model, $g_f \in G$ satisfies that the maximal deviation, $|f(x) - g_f(x)|$ is bounded with high probability within a localized/windowed domain $x \in \mathcal{X}_{\text{window}}$

$$\mathbb{P}\left[\sup_{x\in\mathcal{X}_{\text{window}}} \|f(x) - g_f(x)\| \leq \epsilon\right] \geq 1 - \delta$$

Conjecture

1.1 By considering "locality" as a form of windowing, the spectrum of the problem may be reduced in the local setting in the case Fourier based surrogates, leading to a corresponding reduction in sample complexity.

1.2 By extension any other arbitrary local surrogates should also demonstrate a reduction in sample complexity which should be quantifiable with appropriate metrics.

Numerical Observations

1. Upon fitting an arbitrary signal with a fourier based local/windowed surrogate, periodicity of the signal within the windowed domain is learned with high accuracy.
2. For adequetly wide windows the required frequency spectrum for the Fouriers local surrogate is lowered.

Resource Cost Optimization

A consequence of our conjecture regarding the reduced frequency spectrum is fewer computational resources in form of re-uploading layers would be required for re-training purposes.

Interpretability

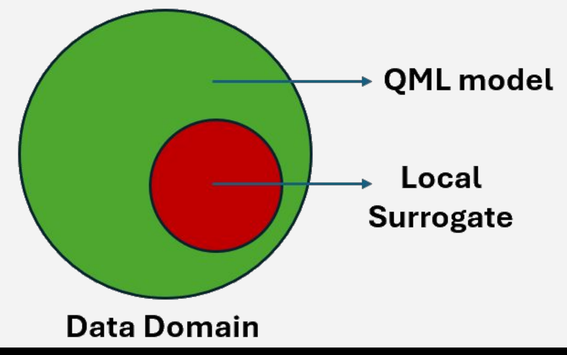
Surrogates are more explainable from the point of digital signal processing. This is because a learning problem has been morphed into fourier analysis problem.

Classical inference

Widely held conjecture regarding the purported quantum advantage in quantum learning models be during the training phase rather than the inference phase means that a local surrogate of a quantum learning model is adequate to perform subsequent inference calls within the windowed domain.

Open problems

1. Mathematical formalism for the concept of local surrogates.
2. Generalized metrics for adequately quantifying equivalence between surrogates (local and global) and quantum learning models.
3. Going beyond Fourier based local surrogates.



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